GB Wind Forecast & Procurement Analysis — January 2026

Prepared for: UK electricity utility · Period: 1–31 Jan 2026 (UTC = UK time) · Data: Elexon BMRS (official)

> **Labels used throughout. [V] Verified** = computed directly from official Elexon data. **[E] Estimate** = derived using a stated assumption. **[N/A]** = needs data we don't have, so not reported.

1. Executive summary

- **The day-ahead wind forecast over-predicted by about 16%.** At the 09:20 day-before deadline, the national forecast promised **~9.99 TWh** of wind against **8.62 TWh** delivered. That is an over-forecast of **~1.38 TWh** [E; hourly forecasts assumed to apply to both half-hours]. On the 744 hours where forecast and actual line up exactly, the gap is the same **+16.0%** [V].
- **The error was a steady lean, not random noise.** Average miss (MAE) was **2,128 MW**, and the average signed error (bias) was **+1,852 MW**. So 87% of the average miss was the forecast running too high [V]. Only 4 of 31 days had a slightly low forecast.
- **The worst days were 28, 12, 11, 1 and 21 January**, with daily MAE of 3.7–5.3 GW. The single worst hours were **00:00–04:00 on 12 January**, when the forecast was **~6.9 GW too high** [V].
- **A later deadline helps only a little.** Moving from 09:20 to 15:30 cuts MAE by 7.5% (2,128 to 1,968 MW). The bias barely changes (+1,852 to +1,817 MW) [V].
- **Money:** buying GB demand at day-ahead prices would have cost about **£2.44 bn** [E]. Covering the shortfall caused by the wind error at imbalance prices would have cost **~£145 m**. **Compared with buying that same energy day-ahead, the extra cost was only ~£4 m net**, because in January imbalance prices did not move with the error (correlation 0.02) [E].
- **We cannot verify the utility's own purchases or costs.** No contract, trade or settlement data was supplied, so those figures are **[N/A]**. Every procurement and cost figure here is a **national proxy**, not this utility's book.

2. Data sources & methodology

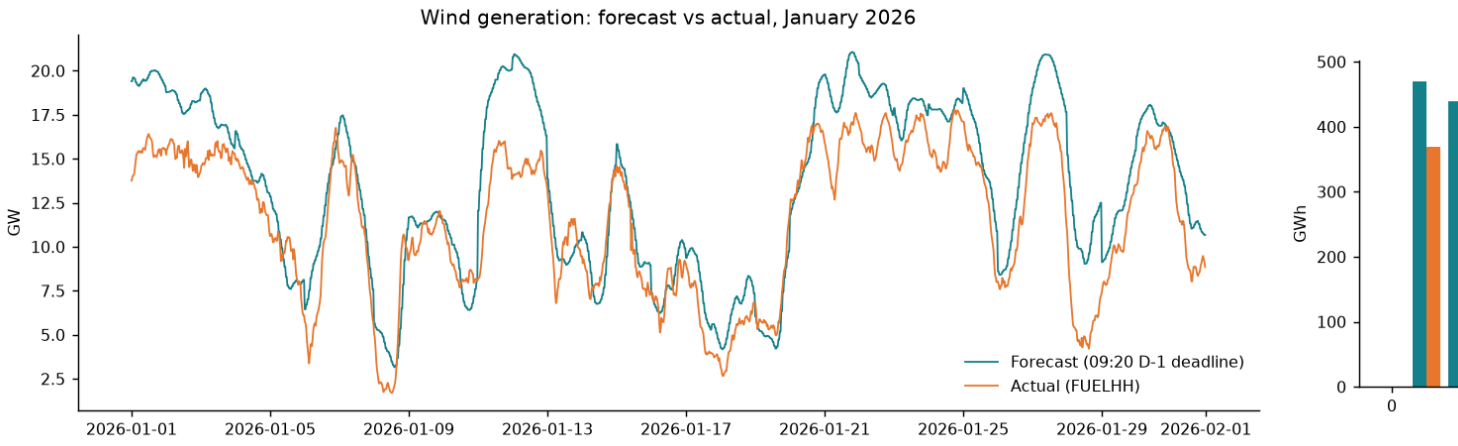
Dataset	Elexon BMRS source	Used for	Rows
Wind forecast (WINDFOR)	`/datasets/WINDFOR/stream` (by publish time, 24 Dec–31 Jan)	Forecast versions	22,776 (0 duplicates, 0 conflicts)
Wind actual (FUELHH, fuelType=WIND)	`/datasets/FUELHH/stream`	Actual generation	1,488 / 1,488 half-hours
National demand (INDO)	`/demand/outturn`	Demand / procurement proxy	1,488 / 1,488
Day-ahead price (Market Index, APX)	`/datasets/MID/stream`	Wholesale price	1,488 half-hours (N2EX index had ~zero volume, excluded)
Imbalance price (SSP/SBP)	`/balancing/settlement/system-prices`	Balancing price	1,488 (single price: SSP = SBP)

Choosing the forecast. For each target hour we use the newest forecast published **at or before** the deadline, which is 09:20 UK time the day before delivery. Forecast runs are published at 02:30, 04:30, 07:30, 09:30, 11:30, 15:30, 18:30 and 22:30 UTC. So the **09:20 deadline always uses the 07:30 run**, 09:50 uses the 09:30 run, and 15:30 uses the 15:30 run. All three are therefore different forecast versions [V]. Average lead time at 09:20 is 28 hours, and the longest lead time seen was 66.5 hours.

Resolution. WINDFOR forecasts are **hourly** (every value falls on :00), while actuals are half-hourly. Accuracy metrics use only the **744 on-the-hour half-hours** where both exist [V]. For monthly energy totals, each hourly forecast is copied to the following :30 half-hour [E].

Metrics (error = forecast - actual, in MW): MAE = mean of the absolute errors. RMSE = square root of the mean squared error. Bias = mean of the signed errors. P90 = 90th percentile of the absolute errors (nearest-rank method). % error = MAE ÷ mean actual (this equals WAPE here). Energy in MWh = sum of MW × 0.5 per half-hour.

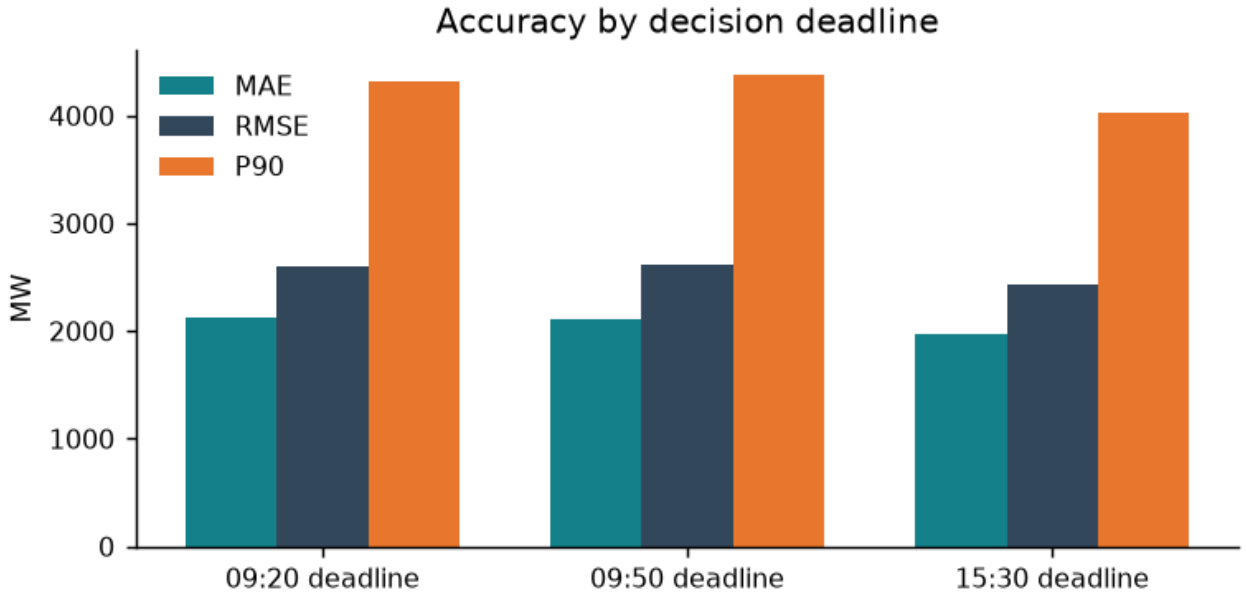
3. Forecast vs actual



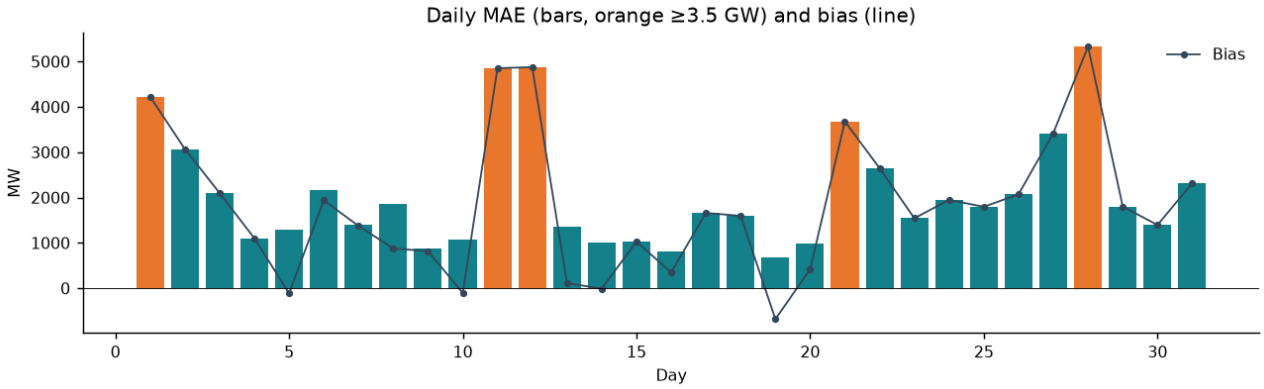
	Forecast	Actual	Difference
Full month energy [E]	9,993,885 MWh	8,616,069 MWh	**+1,377,817 MWh (+16.0%)**
Matched hours only (744 × 0.5 h) [V]	4,996,943 MWh	4,308,174 MWh	+688,769 MWh (+16.0%)
Under-delivery: half-hours where forecast was above actual [E]			1,486,910 MWh
Over-delivery: half-hours where actual was above forecast [E]			109,094 MWh

4. Forecast accuracy [V]

Deadline (D-1)	Run used	MAE (MW)	RMSE (MW)	Bias (MW)	P90 (MW)	% error	Pairs
09:20 (default)	07:30	**2,128**	**2,606**	**+1,852**	**4,313**	**18.4%**	744
09:50	09:30	2,107	2,611	+1,840	4,383	18.2%	744
15:30	15:30	1,968	2,435	+1,817	4,030	17.0%	744



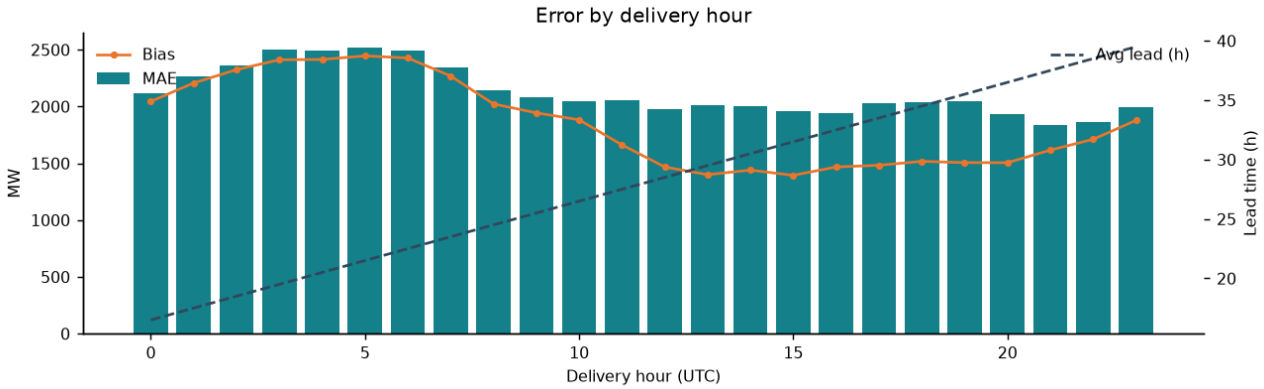
5. Daily and hourly analysis [V]



Largest-error days (09:20 deadline):

Day	MAE (MW)	Bias (MW)	Mean forecast	Mean actual
28 Jan	5,329	+5,329	11,135	5,806
12 Jan	4,875	+4,875	19,320	14,445
11 Jan	4,853	+4,853	18,359	13,506
1 Jan	4,223	+4,223	19,580	15,356
21 Jan	3,677	+3,677	19,390	15,713

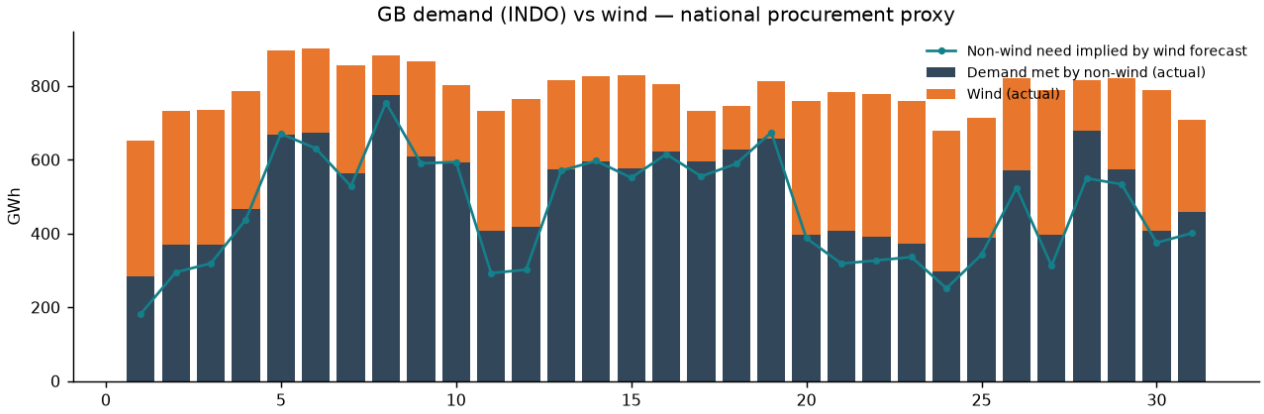
On each of these days MAE equals bias, so the forecast was too high in **every** hour of the day. On 28 Jan, almost twice as much wind was forecast as was delivered.



By time of day: errors peak **overnight, 03:00–06:00 UTC (MAE ~2,500 MW, bias ~+2,430 MW)**. They are lowest at 21:00–22:00 (~1,850 MW). Error does **not** grow with lead time: the delivery hours furthest ahead (evening, ~37 h lead) are among the most accurate. This points to a structural cause of the error rather than one driven by how far ahead the forecast was made.

6. Energy procurement

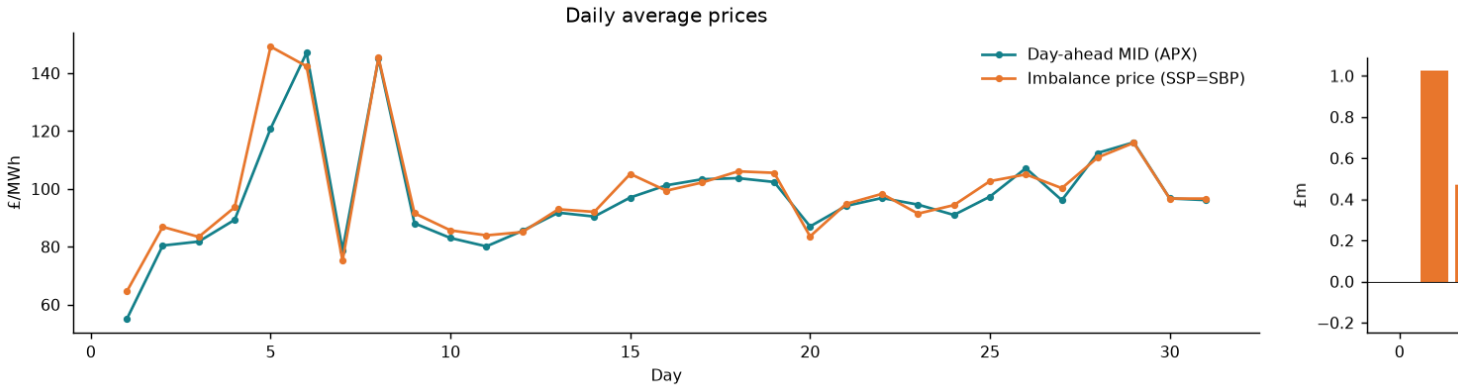
Item	Value	Status
Utility's purchased volume / contracts / position	—	**[N/A]: no trade or settlement data supplied**
GB national demand (INDO)	24.40 TWh	[V]
Wind share of demand (actual)	35.3%	[V]
Energy traded on the APX day-ahead index	4.22 TWh (£436.8 m)	[V], whole market, not this utility
Extra non-wind energy needed because wind under-delivered (national)	~1.49 TWh short, ~0.11 TWh surplus	[E]



How to read the chart: if the buyer had planned non-wind purchases around the 09:20 wind forecast (the teal line), they would have been **short on 27 of 31 days**. Over the month, the extra non-wind energy needed would have been ~1.38 TWh net [E, national scale].

7. Financial analysis [E unless stated]

Item	Value
Average day-ahead price (simple mean) [V]	£97.09/MWh
Average day-ahead price (volume-weighted) [V]	£103.60/MWh
Average imbalance price [V]	£99.35/MWh (peak £750/MWh at 19:00 on 5 Jan)
Indicative cost of all GB demand at day-ahead prices	£2.44 bn (~ £100/MWh)
Cost of covering wind shortfall at imbalance prices	£145.4 m
Value of surplus wind sold at imbalance prices	£11.4 m
Net extra cost of balancing the error instead of buying day-ahead	**~ £4.0 m**
Utility's actual purchase cost / average price / balancing charges	**[N/A]**



How to read this: the **volume** risk is large (~1.4 TWh of wind that never arrived). The **price** penalty for covering it in balancing rather than day-ahead was small in January, because imbalance prices averaged close to day-ahead prices and did not move with the forecast error. The largest single-day hit was **1 January (~£1.0 m)**. This is a national hypothetical: a utility's own exposure scales with its share of the wind position.

8. Key findings

1. **Systematic over-forecast (+1.85 GW, 16%).** The most likely cause is that WINDFOR estimates unconstrained wind potential, while FUELHH records metered output after curtailment and grid constraints. That would also explain why errors are worst overnight on high-wind days, when curtailment is typically highest. *This is a hypothesis we have not verified; it needs BOA/curtailment data to confirm.*

2. **The error was steady, not random.** Most of the average miss was a constant lean in one direction, so a simple correction for that lean could remove much of it (see recommendations).
3. **Event days dominate.** Five days produce a large share of the total error, and all of them are high-wind or sharp wind-drop days (for example, 28 Jan: 11.1 GW forecast vs 5.8 GW delivered).
4. **The deadline choice matters only a little.** 09:20, 09:50 and 15:30 use different forecast runs, but 15:30 reduces MAE by only ~160 MW.
5. **Price risk was muted this month, but tail risk is real.** Imbalance prices reached £750/MWh on 5 Jan. A shortfall during such a spike would cost many times the monthly average.

9. Recommendations

1. **Correct the forecast for its lean before buying.** Subtract a rolling bias (e.g. the past 7–14 days' signed error, varying by hour of day) from the forecast before setting purchases. Back-test it on Feb–Mar data before relying on it.
2. **Buy more non-wind energy ahead of time on high-wind nights**, when over-forecast risk is greatest (00:00–07:00 UTC, forecast above ~15 GW).
3. **Use the 15:30 run for any volume that can still be adjusted** in the 30-minute auction, and keep 09:20 for the hourly day-ahead block.
4. **Obtain the missing data** so these estimates can become verified results: the utility's trades and contracts, its settlement (Elexon SAA/BSC) statements, and curtailment/BOA volumes.
5. **Repeat this monthly**, and add a planning range built from past errors only (P10–P90), so buyers see risk, not just a single number.

10. Limitations

- National, wind-only data for one month. The finance figures are national hypotheticals and do **not** describe this utility.
- Forecast and actual may measure slightly different things (unconstrained forecast vs metered, post-curtailment output; embedded wind treatment may also differ).
- Monthly energy totals assume each hourly forecast also applies to the following half-hour.
- The analysis looks back over January; it is not a forecast of future error.